INCH-POUND

MIL-DTL-32411A
w/AMENDMENT 2
15 November 2023
SUPERSEDING
MIL-DTL-32411A
w/AMENDMENT 1
31 March 2021

DETAIL SPECIFICATION

# IGNITER, TIME BLASTING FUSE: WITH SHOCK TUBE CAPABILITY - M81, LOAD, ASSEMBLE, AND PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense within the distribution limitations noted at the bottom of the page.

## 1. SCOPE

1.1 Scope. This specification describes the requirements and verification methods for the following item: IGNITER, TIME BLASTING FUSE: with shock tube capability, M81.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM-AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERICAL ITEM DESCRIPTION

| | | |
|---|---|---|
| A-A-208B | - | Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces) |
| A-A-59383 | - | Marking Media, Opaque (Porous and Non-Porous Surfaces) |

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-45144 | - | Fuse, Blasting, Time- M700, and Dummy Fuse, Blasting, Time - Inert Loading, Assembly and Packing |
| MIL-P-20444 | - | Primer, Percussion, M42 Parts for and Loading, Assembling and Packing |

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-1168 | - | Ammunition Lot Numbering and Ammunition Data Card |
| MIL-STD-1916 | - | DOD Preferred Methods for Acceptance of Products |
| MIL-STD-2073-1 | - | Standard Practice for Military Packaging |

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US ARMY CCDC - AC DRAWINGS

| | | |
|---|---|---|
| 7553347 | - | Box, Wirebound Boxes, Ammunition, M2A2 |
| 8822498 | - | Cap, Fuse Holder |
| 8822499 | - | Collet |
| 8822500 | - | Grommet |
| 8822502 | - | Washer, Large |
| 8822503 | - | Washer, Small |
| 8822504 | - | Firing Assembly |
| 8822505 | - | Cap, Top |
| 8822506 | - | Housing |
| 8822507 | - | Ring, Pull |



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| | | |
|---|---|---|
| 8822508 | - | Rod, Pull |
| 8822509 | - | Spring, Firing Pin |
| 8822510 | - | Washer, Friction Seal |
| 8822511 | - | Washer, Release |
| 8822512 | - | Striker Assembly |
| 8822513 | - | Housing, Firing Pin |
| 8822514 | - | Pin, Firing |
| 9345327 | - | Primer, Alternate |
| 12972588 | - | Base, Primer |
| 12972589 | - | Plug, Weatherproofing |
| 12972590 | - | Plug, Shipping |
| 12972591 | - | Primer, Base Igniter |
| 12972620 | - | Igniter, Time Blasting Fuse: With Shock Tube Capability, M81 |
| 12977189 | - | Box, Set-Up |
| 12977190 | - | Bag, Barrier |
| 12977191 | - | Box, Wood |
| 12977192 | - | Packing & Marking for Box, Set-Up For Igniter, Time Blasting Fuse w/ Shock Tube Capability M81 |
| 12977193 | - | Packing & Marking for Box, Wood For Igniter, Time Blasting Fuse w/ Shock Tube Capability M81 |
| 13019660 | - | Packing and Marking for 54 Igniter, Time Blasting Fuse, Weatherproof, M81 |
| 13019661 | - | Packing and Marking for 108 Igniter, Time Blasting Fuse, Weatherproof, M81 |
| 13026126 | - | Shock tube |
| 13032542 | - | Box, Ammunition, M2A2 Assembly |
| 13042427 | - | Tray |
| 13051505 | - | Pin, Cotter |

(Copies of the above drawings are available by email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil.)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

| | | |
|---|---|---|
| 19-48-4116 | - | Basic Procedures: Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets |
| 19-48-4116/131B | - | Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Igniter, Time Blasting Fuse, M81, Packed 108 per Wirebound Box, Unitized 48 Boxes per 40” x 48” Pallet; Approx Box size 14-7/16” L x 12-17/32” W x 8-1/8” H |
| 19-48-4116/131D | - | Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Igniter, Time Blasting Fuse, Packed 360 per Wooden Box, Unitized |



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24 Boxes per 40” x 48” Pallet; Approx Box size 21-1/2” L x 12” W x 14-1/2” H

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Topics/Drawings.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM B117 | - | Standard Practice for Operating Salt Spray (Fog) Apparatus |
| ASTM F519 | - | Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Coating Processes and Service Environments |

(Copies of the above documents are available online at https://astm.org .)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Materials, components, and assemblies. All materials, components and assemblies shall comply with the requirements of Drawing 12972620 and its subsidiary drawings.

3.2.1 Hydrogen embrittlement. The firing pin spring and pull ring shall be free of embrittlement.

3.2.2 Salt spray. The firing pin spring, pull ring, and cotter pin shall be subjected to the required salt spray test as specified on the applicable drawing to determine compliance. Except at the edge of an unprotected area, the salt spray tested samples shall show no evidence of corrosion products at the end of the 96 hour test.



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3.3 Time fuse initiation. The igniter shall initiate M700 Time Blasting Fuse conforming to MIL-DTL-45144.

3.4 Time fuse attachment integrity. The igniter shall not separate from M700 Time Blasting Fuse, conforming to MIL-DTL-45144, after exposure to not less than 25 pounds (lbs) of axial force for not less than 10 seconds. After this exposure, it shall initiate the time fuse.

3.5 Shock tube initiation. The igniter shall initiate the shock tube conforming to Drawing 13026126-3.

3.6 Shock tube attachment integrity. The igniter shall not separate from shock tube, conforming to Drawing 13026126-3, after exposure to not less than 5 pounds (lbs) of axial force for not less than 10 seconds. After this exposure, it shall initiate the shock tube.

3.7 Operating temperatures. The igniter shall operate throughout the temperature range of -50ºF to 130ºF.

3.8 Immersion. The igniter shall operate after immersion in water at a depth not less than 30 inches for not less than 6 hours.

3.9 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.10 Workmanship. All parts, sub-assemblies, and assemblies shall be clean and free of burrs, chips, sharp edges, cracks, unblended radii, burn marks, checks, sinks, crazes, blisters, excess flash, dirt, grease, oil, rust, corrosion, and other foreign matter.



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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

| Method of Verification<br>1 – Analysis 3 – Examination<br>2 – Demonstration 4 – Test | | Classes of Verification<br>A – First article<br>B – Conformance | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Section 3 and 5 Requirements | Description | Verification Method | | | | Verification Class | | Section 4 Verification |
| | | 1 | 2 | 3 | 4 | A | B | |
| 3.1.1 | First article | | X | X | X | X | | 4.2 |
| 3.1.2 | Conformance inspection | | X | X | X | | X | 4.3 |
| 3.2 | Materials, components, and assemblies | | | X | X | X | X | 4.3.4.1 – 4.3.4.23 |
| 3.2.1 | Hydrogen embrittlement | | | | X | X | X | 4.5.4 |
| 3.2.2 | Salt spray | | | | X | X | X | 4.5.5 |
| 3.3 | Time fuse initiation | | | | X | X | X | 4.5.6 – 4.5.8 |
| 3.4 | Time fuse attachment integrity | | | | X | X | X | 4.5.6 |
| 3.5 | Shock tube initiation | | | | X | X | X | 4.5.6 – 4.5.8 |
| 3.6 | Shock tube attachment integrity | | | | X | X | X | 4.5.6 |
| 3.7 | Operating temperatures | | | | X | X | X | 4.5.7 |
| 3.8 | Immersion | | | | X | X | X | 4.5.8 |
| 3.9 | Ammunition lot numbering | | | X | | X | X | 4.3.3 |
| 3.10 | Workmanship | | | X | | X | X | 4.3.4.1 – 4.3.4.32 |
| 5.1 | Packaging and marking | | | X | X | X | X | 4.3.4.24- 4.3.4.32 |
| 5.1.1 | Barrier bag resistance to leakage | | | | X | X | X | 4.3.4.25& 4.6.1, 4.6.2 |
| 5.1.2 | Marking ink/media | | X | X | | X | X | 4.3.4.25& 4.3.4.27 - 4.3.4.32, 4.6.4 |

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (See Table I):

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures, and facilities as will be used during regular production.



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4.2.1 First article rejection. If one (1) or more assemblies, subassemblies, materials, or components fail to comply with the first article requirements, the sample(s) and first article shall be rejected.

TABLE II. First article tests and inspections.

| Examination or test | Drawing(s) | Sample Size | Requirement Paragraph | Paragraph Reference / Inspection Method |
|---|---|---|---|---|
| Ring, pull | 8822507 | | | |
| Examination for defects | | 25 | 3.2 | 4.3.4.1 |
| Hydrogen embrittlement | | 4.5.4 | 3.2.1 | 4.5.4 |
| Salt spray | | 5 | 3.2.2 | 4.5.5 |
| Rod, pull | 8822508 | 25 | 3.2 | 4.3.4.2 |
| Spring, firing pin | 8822509 | | | |
| Examination for defects | | 25 | 3.2 | 4.3.4.3 |
| Hydrogen embrittlement | | 4.5.4 | 3.2.1 | 4.5.4 |
| Salt spray | | 5 | 3.2.2 | 4.5.5 |
| Housing, firing pin | 8822513 | 25 | 3.2 | 4.3.4.4 |
| Pin, firing | 8822514 | 25 | 3.2 | 4.3.4.5 |
| Striker assembly | 8822512 | 25 | 3.2 | 4.3.4.6 |
| Housing | 8822506 | 25 1/ | 3.2 | 4.3.4.7 |
| Cap, top | 8822505 | 25 1/ | 3.2 | 4.3.4.8 |
| Pull rod and striker assembly | 8822508 & 8822512 | 80 | 3.2 | 4.3.4.9 |
| Firing assembly | 8822504 | 25 | 3.2 | 4.3.4.10 |
| Washer, friction seal | 8822510 | 25 1/ | 3.2 | 4.3.4.11 |
| Cap, fuse holder | 8822498 | 25 1/ | 3.2 | 4.3.4.12 |
| Collet | 8822499 | 25 1/ | 3.2 | 4.3.4.13 |
| Grommet | 8822500 | 25 1/ | 3.2 | 4.3.4.14 |
| Washer, release | 8822511 | 25 1/ | 3.2 | 4.3.4.15 |
| Plug, weatherproofing | 12972589 | 25 1/ | 3.2 | 4.3.4.16 |
| Plug, shipping | 12972590 | 25 1/ | 3.2 | 4.3.4.17 |
| Base, primer | 12972588 | 25 1/ | 3.2 | 4.3.4.18 |
| Primer, alternate | 9345327 | | | |
| Examination for defects | | 25 | 3.2 | 4.3.4.19 |
| Sensitivity test | | 4.3.4.18 | 3.2 | 4.3.4.19 |
| Dud test | | 4.3.4.18 | 3.2 | 4.3.4.19 |
| Primer, base igniter subassembly | 12972591 | 25 1/ | 3.2 | 4.3.4.20 |
| Washer, large | 8822502 | 25 1/ | 3.2 | 4/ |
| Washer, small | 8822503 | 25 1/ | 3.2 | 4/ |
| Pin, cotter | 13051505 | | | |
| Examination for defects | | 25 | 3.2 | 4.3.4.21 |
| Salt spray | | 5 | 3.2.2 | 4.5.5 |
| M81 Igniter | 12972620 | | | |
| Examination for defects | | 100 2/ | 3.2 | 4.3.4.22 |



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TABLE II. First article tests and inspections (continued)

| Examination or test | Drawing(s) | Sample Size | Requirement Paragraph | Paragraph Reference / Inspection Method |
|---|---|---|---|---|
| Cotter pin extraction | | 20 3/ | 3.2 | 4.5.1 |
| Pull ring force limits | | 3/ | 3.2 | 4.5.2 |
| Disassembly | | 3/ | 3.2 | 4.5.9 |
| Attachment integrities | | 80 | 3.4, 3.6 | 4.5.6 |
| Operating temperatures | | 120 | 3.7 | 4.5.7 |
| Immersion | | 80 | 3.8 | 4.5.8 |
| Setup box | 12977192 | 7/ | 5.1 | 4.3.4.24 – 4.3.4.25 |
| Barrier bags | 12977190 | | 5.1 | 4.3.4.25 |
| Leakage test | | 5/ | 5.1.1 | 4.6.1 |
| Heat seal seam strength test | | 5/ | 5.1.1 | 4.6.2 |
| M2A2 ammunition can (Marine) | 13019660 | 6/ | 5.1 | 4.3.4.26 – 4.3.4.27 |
| Marking resistance to water | | 5.1.2 | 5.1.2 | 4.6.4, 4.6.4.1 |
| Marking resistance to smearing | | 5.1.2 | 5.1.2 | 4.6.4, 4.6.4.2 |
| Wood box (Army) | 12977193 | 7/ | 5.1 | 4.3.4.28 – 4.3.4.29 |
| Wirebound box (Marine) | 13019661 | 7/ | 5.1 | 4.3.4.30 - 4.3.4.31 |
| Unitized load | | 7/ | 5.1 | 4.3.4.32 |

Notes:

1/ If a multiple cavity mold is used, then not less than five (5) samples from each cavity shall be taken. The sample size of the total multiple cavity quantities shall be not less than the quantity specified In Table II. Cavity identification shall be maintained to permit tracing each part to the cavity in which it was molded.

2/ After examination for defects, reuse and subject the igniters to FAT tests.

3/ The same igniters shall undergo cotter pin extraction testing followed by pull ring force limits testing followed by disassembly.

4/ Verify all dimensions on the drawing.

5/ The test shall be performed on three (3) sealed barrier bags from each sealer.

6/ The tests shall be performed on two (2) marked M2A2 cans (Marine).

7/ For packaging: The examinations shall be performed on one (1) full pack and one (1) light pack.

For light packs: The light pack shall be no less than 50% filled and the remainder packed out. The light pack instructions are found on the applicable packing and marking drawing.

For unitized loads: The examinations shall be performed on one (1) sample.

Packaging within unitized load or assembled packaging may include samples from lower level packaging when conformance can still be demonstrated at all levels. With prior Government approval, the M81s to be enclosed in the packaging may be substituted with inert mass and volume analogues, reused for each sample, or excluded from packaging if not necessary to demonstrate conformance.



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4.3 Conformance inspection.

4.3.1 Conformance rejection. If any assembly, subassembly, material, or component fails to comply with the conformance inspection requirements of 4.3 and its subparagraphs, the lot from which the sample was taken shall be rejected.

4.3.2 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specification and revisions. In addition each lot shall contain materials from not more than one lot interfix number from one manufacturer except the following:

a. Primers from not more than one lot from one manufacturer.
b. Primers shall be no older than 5 years from the time of manufacturing.

4.3.3 Ammunition lot number. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.4 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.31, the definitions of critical, major, and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).



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| 4.3.4.1 Ring, pull. | | | Drawing Number 8822507<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Hydrogen embrittlement | 4.5.4 | 3.2.1 | 4.5.4 |
| 102 | Salt spray | 5 | 3.2.2 | 4.5.5 |
| | | | | |
| Minor | | | | |
| 201 | Protective finish damaged, missing, or incomplete | VL II | 3.2 | Visual |
| 202 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.2 Rod, pull. | | | Drawing Number 8822508 | |
|---|---|---|---|---|
| | | | Next Higher Assembly 8822504 | |
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Diameter at front end (.186 +.000/-.002) | VL IV | 3.2 | Gage |
| 102 | Perpendicularity of shoulder with diameter at rear end (.002) | VL IV | 3.2 | Gage |
| 103 | Radius of shoulder (.002 max) | VL IV | 3.2 | Gage |
| 104 | True position of safety pin hole | VL IV | 3.2 | Gage |
| 105 | Pressure relief cavity or hole incomplete or blocked | VL IV | 3.2 | Visual |
| 106 | Diameter of cotter pin hole (.078 +.004/- .001) | VL IV | 3.2 | Gage |
| 107 | Diameter of relief cavity (.078 +.004/-.001) | VL IV | 3.2 | Gage |
| 108 | Depth of relief cavity (1.000 +/-.010) | VL IV | 3.2 | Gage |
| 109 | Diameter of relief hole (.078 +.004/-.001) | VL IV | 3.2 | Gage |
| 110 | True position of relief hole | VL IV | 3.2 | Gage |
| 111 | Small diameter of rod (.149 +.000/ -.003) | VL IV | 3.2 | Gage |
| 112 | Small diameter rod surface finish (63 microns) | VL IV | 3.2 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Total length (2.17 +.00/-.02) | VL II | 3.2 | Gage |
| 202 | Finish damaged, missing, or incomplete | VL II | 3.2 | Visual |
| 203 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.3 Spring, firing pin. | | | Drawing Number 8822509<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Hydrogen embrittlement | 4.5.4 | 3.2.1 | 4.5.4 |
| 102 | Solid height (.84 max) | VL IV | 3.2 | Gage |
| 103 | Load at .89" height (4.2 +/-.4 lbs) | VL IV | 3.2 | Gage |
| 104 | Protective finish damaged, missing, or incomplete | VL IV | 3.2 | Visual |
| 105 | End not squared or ground | VL IV | 3.2 | Visual |
| 106 | Salt spray | 5 | 3.2.2 | 4.5.5 |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.4 Housing, firing pin. | | | Drawing Number 8822513<br>Next Higher Assembly 8822512 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Perpendicularity of cavity base with inside diameter at front end (.002) | VL IV | 3.2 | Gage |
| 102 | Outside diameter (.219+.000/-.003) | VL IV | 3.2 | Gage |
| 103 | Inside diameter at front end (.187 +.002/-.000) | VL IV | 3.2 | Gage |
| 104 | Total length (.89 +.00/-.01) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Finish damaged, missing, or incomplete | VL II | 3.2 | Visual |
| 202 | Presence of slots (4x) | VL II | 3.2 | Visual |
| 203 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.5 Pin, firing. | | | Drawing Number 8822514<br>Next Higher Assembly 8822512 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Total length (.45 +.00/-.01) | VL IV | 3.2 | Gage |
| 102 | Length of point (.100 +.015/-.000) | VL IV | 3.2 | Gage |
| 103 | Profile of point (spherical .08 +.00/-.01) | VL IV | 3.2 | Gage |
| 104 | Location of crimp groove (.13 +00/-.01) | VL IV | 3.2 | Gage |
| 105 | Hole diameter (.054 +.005/-.000, x10) | VL IV | 3.2 | Gage |
| 106 | .110 min from center of part to inside edge of hole (x10) | VL IV | 3.2 | Gage / Visual |
| 107 | Outside diameter not breached or compromised by holes | VL IV | 3.2 | Visual |
| 108 | True position of flange diameter | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Distance from base to flange (.25 +.01/-.00) | VL II | 3.2 | Gage |
| 202 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.6 Striker assembly. | | | Drawing Number 8822512<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Perpendicularity of forward face of flange with outside diameter of firing pin housing (.003) | VL IV | 3.2 | Gage |
| 102 | Total length (1.10 max) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Crimp improper (shallow or less than 360°) | VL II | 3.2 | Visual |
| 202 | Holes not obstructed | VL II | 3.2 | Visual |
| 203 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.7 Housing. | | | Drawing Number 8822506<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Major diameter of thread (5/8-20 UN-2A, x2) | VL IV | 3.2 | Gage |
| 102 | Pitch diameter of thread (5/8-20 UN-2A, x2) | VL IV | 3.2 | Gage |
| 103 | Inside diameter (.39 +.01/-.00) | VL IV | 3.2 | Gage |
| 104 | Total length (3.18 +.00/-.03) | VL IV | 3.2 | Gage |
| 105 | Threads damaged | VL IV | 3.2 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Outside diameter (.627 +.000/-.010) | VL II | 3.2 | Gage |
| 202 | Perpendicularity of closed end surface | VL II | 3.2 | Gage |
| 203 | Diameter of pull rod hole (.150 +.005/-.000) | VL II | 3.2 | Gage |
| 204 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.8 Cap, top. | | | Drawing Number 8822505<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Distance from front end to bottom of thread (.56 +.05/-.00) | VL IV | 3.2 | Gage |
| 102 | Total length (.95 +.00/-.02) | VL IV | 3.2 | Gage |
| 103 | Pitch diameter of thread (5/8-20 UN-2B) | VL IV | 3.2 | Gage |
| 104 | Minor diameter of thread (5/8-20 UN-2B) | VL IV | 3.2 | Gage |
| 105 | Location of cotter pin hole (.80 +/-.01) | VL IV | 3.2 | Gage |
| 106 | Crack or split | VL IV | 3.2 | Visual |
| 107 | Surface in bottom not smooth or free of flash | VL IV | 3.2 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Diameter of pull rod hole (.150 +.005/-.000) | VL II | 3.2 | Gage |
| 202 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.9 Subassembly of pull rod and striker assembly. | | | Drawing Number 8822508 & 8822512<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Firing pin load test | 1/ | 3.2 | 4.5.3 |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | 1/ | 3.10 | Visual |

Notes:
1/ Test 50 subassemblies of the pull rods and striker assemblies per igniter lot.

If one (1) or more subassemblies fail to comply with the specified requirement, then the M81, pull rod, firing pin, and striker assembly lots shall be rejected.



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| 4.3.4.10 Firing assembly. | | | Drawing Number 8822504<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Pull rod not engaged with striker assembly | VL IV | 3.2 | Manual |
| 102 | Depth from fuse holder end to firing pin flange (1.12 +.06/-.00) | VL IV | 3.2 | Gage |
| 103 | External components missing or loose (pull ring, top cap, cotter pin with cord, pull rod, striker assembly) | VL IV | 3.2 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Cord improperly tied | VL II | 3.2 | Visual |
| 202 | Threads damaged | VL II | 3.2 | Visual |
| 203 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.11 Washer, friction seal. | | | Drawing Number 8822510<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Outside diameter (.56 +.00/-.02) | VL IV | 3.2 | Gage |
| 102 | Inside diameter (.080 +.015/-.000) | VL IV | 3.2 | Gage |
| 103 | Thickness (.075 +.000/-.010) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.12 Cap, fuse holder. | | | Drawing Number 8822498<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Pitch diameter of thread (5/8-20 UN-2B) | VL IV | 3.2 | Gage |
| 102 | Minor diameter of thread (5/8-20 UN-2B) | VL IV | 3.2 | Gage |
| 103 | Total length (1.10 +.00/-.03) | VL IV | 3.2 | Gage |
| 104 | Distance from rear end to cavity below thread (.78 +.02/-.00) | VL IV | 3.2 | Gage |
| 105 | Distance from rear end to bottom of thread (.63 +.05/-.00) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.13 Collet. | | | Drawing Number 8822499<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Inside radius (.095 +.005/-.000) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.14 Grommet. | | | Drawing Number 8822500<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Inside diameter (.212 +.005/-.000) | VL IV | 3.2 | Gage |
| 102 | Height (.314 +.000/-.005) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.15 Washer, release. | | | Drawing Number 8822511<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Outside diameter of largest diameter (.38 +.000/-.02 | VL IV | 3.2 | Gage |
| 102 | Outside diameter of smallest diameter (.205 +.000/-.005) | VL IV | 3.2 | Gage |
| 103 | Inside diameter (.152 +.005/-.000) | VL IV | 3.2 | Gage |
| 104 | Thickness (.03 +.00/-.01) | VL IV | 3.2 | Gage |
| 105 | Chamfer (.015 +.005/-.000 x 45 Degrees) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.16 Plug, weatherproofing. | | | Drawing Number 12972589<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Outside diameter of smallest diameter (.118 +.000/-.005) | VL IV | 3.2 | Gage |
| 102 | Length of smallest diameter (1.00 +.00/-.03) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Overall length (1.61 +.00/-.03) | VL II | 3.2 | Gage |
| 202 | Workmanship | VL II | 3.10 | Visual |

| 4.3.4.17 Plug, shipping. | | | Drawing Number 12972590<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Inside diameter (.118 +.005/-.000) | VL IV | 3.2 | Gage |
| 102 | Overall length (1.00 +.00/-.03) | VL IV | 3.2 | Gage |
| 103 | Outside diameter of small outside diameter (.212 +.000/-.005) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Outside diameter of knob (.625 +.000/-.020) | VL II | 3.2 | Gage |
| 202 | Thickness of knob (.12 +.00/-.02) | VL II | 3.2 | Gage |
| 203 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.18 Base, primer. | | | Drawing Number 12972588 | |
|---|---|---|---|---|
| | | | Next Higher Assembly 12972591 | |
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Overall length (.51 +.00/-.01) | VL IV | 3.2 | Gage |
| 102 | Inside diameter of small diameter hole (.100 +.005/-.000) | VL IV | 3.2 | Gage |
| 103 | Inside diameter of counter bore (.1715 +.0030/-.0000) | VL IV | 3.2 | Gage |
| 104 | Length of shoulder from slotted end (.130 +.000/-.005) | VL IV | 3.2 | Gage |
| 105 | Depth of counter bore (.116 +.008/- .000) | VL IV | 3.2 | Gage |
| 106 | Outside diameter of large diameter (.445 +.000/-.005) | VL IV | 3.2 | Gage |
| 107 | Outside diameter of small diameter (.373 +.000/-.005) | VL IV | 3.2 | Gage |
| 108 | True position of 5 longitudinal slots relative to alignment | VL IV | 3.2 | Gage |
| 109 | Depth of 5 longitudinal slots (.060 +.005/-.000) | VL IV | 3.2 | Gage |
| 110 | True position of longitudinal alignment key | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.19 Primer, alternate. | | | Drawing Number 9345327<br>Next Higher Assembly 12972591 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Height (.115 +/- .004) | VL IV | 3.2 | Gage |
| 102 | Diameter (.1752 +/- .0004) | VL IV | 3.2 | Gage |
| 103 | Sensitivity testing | 1/ | 3.2 | 1/ |
| 104 | Dud testing | 2/ | 3.2 | 2/ |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |
| Notes:<br>1/ Sensitivity testing shall be in accordance with MIL-P-20444 paragraph 4.2.3.2.1 and 9345327.<br>2/ Dud testing shall be in accordance with MIL-P-20444 paragraph 4.2.3.2.2 and 9345327. | | | | |

| 4.3.4.20 Primer, base igniter subassembly. | | | Drawing Number 12972591<br>Next Higher Assembly 12972620 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Primer missing, inverted, or damaged | VL IV | 3.2 | Visual |
| 102 | Primer not fully seated | VL IV | 3.2 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL II | 3.10 | Visual |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:17Z
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| 4.3.4.21 Pin, cotter. | | | Drawing Number 13051505<br>Next Higher Assembly 8822504 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Salt spray | 5 | 3.2.2 | 4.5.5 |
| | | | | |
| Minor | | | | |
| 201 | Protective finish damaged, missing, or incomplete | VL II | 3.2 | Visual |
| 202 | Workmanship | VL II | 3.10 | Visual |



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| 4.3.4.22 Igniter, time blasting fuse: with shock tube capability – examinations. | | | Drawing Number 12972620<br>Next Higher Assembly 12977192 | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria 1/ | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Total length (5.348 max.) | VL IV | 3.2 | Gage |
| 102 | Length from top cap to end of fuse holder cap (4.00 max.) | VL IV | 3.2 | Gage |
| 103 | External components missing or loose (top cap, pull rod, weatherproofing plug, shipping plug, fuse holder cap, cotter pin, pull ring, cord) | VL IV | 3.2 | Visual/ Manual |
| 104 | Top cap, fuse holder cap, or housing cracked or split | VL IV | 3.2 | Visual |
| 105 | Markings missing, illegible or incorrect | VL IV | 5.1.2 | 4.6.4 |
| | | | | |
| Minor | | | | |
| 201 | Cotter pin not spread equally | VL II | 3.2 | Visual |
| 202 | Cotter pin cord missing or improperly tied | VL II | 3.2 | Visual |
| 203 | Protective finish missing, or incomplete (pull ring and cotter pin) | VL II | 3.2 | Visual |
| 204 | Marking color improper | VL II | 3.2 | Visual |
| 205 | Knurl missing (where applicable) | VL II | 3.2 | Visual |
| 206 | Workmanship | VL II | 3.10 | Visual |

Notes:
1/ Sample size shall be in accordance with the MIL-STD-1916 verification level, plus all the destructive test units of 4.3.4.21.



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| 4.3.4.23 Igniter, time blasting fuse: with shock tube capability – destructive. | | | Drawing Number 12972620<br>Next Higher Assembly | |
|---|---|---|---|---|
| | Examinations and Tests | Conformance Criteria 1/ | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Cotter pin extraction | 20 2/ | 3.2 | 4.5.1 |
| 102 | Pull ring force limits | 2/ | 3.2 | 4.5.2 |
| 103 | Attachment integrities | 80 | 3.4, 3.6 | 4.5.6 |
| 104 | Operating temperatures | 120 | 3.7 | 4.5.7 |
| 105 | Immersion | 80 | 3.8 | 4.5.8 |
| 106 | Disassembly | 20 2/ | 3.2 | 4.5.9 |

Notes:
1/ The total function test quantity is 300.

2/ The same sample igniters shall undergo sequential cotter pin extraction testing followed by pull ring force limits testing followed by disassembly.



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| 4.3.4.24 Packing and marking for box, set-up for M81 igniter (prior to sealing). | | | Drawing Number 12977192<br>Next Higher Assembly 13019660 19277193 | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Missing items | VL II | 5.1 | Visual |
| 102 | Packaging material moisture content incorrect | 1/ | 5.1 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Insufficient filler | VL I | 5.1 | 2/ |
| 202 | Workmanship | VL I | 3.10 | Visual |

Notes:
1/ At the start of each production day, three (3) samples of all paper packaging material contained within the barrier bag (setup box) shall be randomly selected and subjected to moisture content analysis.
2/ Visually verify no gaps exist inside the container (causing a loose pack).



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| 4.3.4.25 Packing and marking for box, set-up for M81 igniter (after sealing). | | | Drawing Number 12977192<br>Next Higher Assembly 13019660 19277193 | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Sealing strip torn, badly wrinkled or otherwise fails to seal box completely or correctly | VL II | 5.1 | Visual |
| 102 | Markings missing, incorrect, illegible, incorrect size/height, or not in proper location | VL II | 5.1.2 | 4.6.4 / Gage |
| 103 | Leakage test | 1/ | 5.1.1 | 4.6.1 |
| 104 | Heat seal seam strength test | 1/ | 5.1.1 | 4.6.2 |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL I | 3.10 | Visual |

Notes
1/ The test shall be performed at the beginning of each shift on one (1) sealed barrier bag from each sealer. For sampling purposes, consider a shift to be no more than 12 hours.



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| 4.3.4.26 Packing and marking drawing for the M81 Marine packaging M2A2 ammo can (prior to sealing). | | | Drawing Number 13019660<br>Next Higher Assembly 13019661 | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Missing items | VL II | 5.1 | Visual |
| 102 | Incorrect humidity indicator card or desiccant bag | VL II | 5.1 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Insufficient filler | VL I | 5.1 | 1/ |
| 202 | Workmanship | VL I | 3.10 | Visual |
| Notes<br>1/ Visually verify no gaps exist inside the container (causing a loose pack). | | | | |

| 4.3.4.27 Packing and marking for the M81 Marine M2A2 ammo can (after sealing). | | | Drawing Number 13019660<br>Next Higher Assembly 13019661 | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Metallic seal missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly | VL II | 5.1 | 4.6.3 |
| 102 | Markings missing, incorrect, illegible, incorrect size/height, or not in proper location | VL II | 5.1.2 | 4.6.4/ Gage |
| 103 | Can punctured, dented, cracked, or damaged | VL I | 5.1 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL I | 3.10 | Visual |



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| 4.3.4.28 Packing and marking drawing for the M81 Army wood box (prior to sealing). | | | Drawing Number 12977193<br>Next Higher Assembly | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Missing items | VL II | 5.1 | Visual |
| 102 | Markings – Preservative, Heat Treatment, and Box Manufacturer – missing, incorrect, illegible, incorrect size/height, or not in proper location | VL II | 5.1.2 | Visual/ Gage |
| | | | | |
| Minor | | | | |
| 201 | Insufficient filler | VL I | 5.1 | 1/ |
| 202 | Workmanship | VL I | 3.10 | Visual |
| Notes<br>1/ Visually verify no gaps exist inside the container (causing a loose pack). | | | | |



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| 4.3.4.29 Packing and marking drawing for the M81 Army wood box (after sealing). | | | Drawing Number 12977193<br>Next Higher Assembly 19-48-4116 19-48-4116/131D (M81 Army) | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Box, hinge, or hasp damaged | VL II | 5.1 | Visual |
| 102 | Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly | VL II | 5.1 | 4.6.3 |
| 103 | Markings –including 2D bar codes – missing, incorrect, illegible, incorrect size/height, or not in proper location | VL II | 5.1.2 | 4.6.4/ Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL I | 3.10 | Visual |



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| 4.3.4.30 Packing and marking drawing for the M81 Marine wirebound box overpack (prior to sealing). | | | Drawing Number 13019661<br>Next Higher Assembly | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Missing Items | VL II | 5.1 | Visual |
| 102 | Markings – Preservative, Heat Treatment, and Box Manufacturer- missing, incorrect, illegible, incorrect size/height, or not in proper location | VL II | 5.1.2 | Visual/ Gage |
| | | | | |
| Minor | | | | |
| 201 | Insufficient filler | VL I | 5.1 | 1/ |
| 201 | Workmanship | VL I | 3.10 | Visual |
| Notes<br>1/ Visually verify no gaps exist inside the container (causing a loose pack). | | | | |



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| 4.3.4.31 Packing and marking drawing for the M81 Marine wirebound box overpack (after sealing). | | | Drawing Number 13019661<br>Next Higher Assembly 19-48-4116 19-48-4116/131B (M81 Marine) | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Box, hinge, or hasp damaged | VL II | 5.1 | Visual |
| 102 | Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly | VL II | 5.1 | 4.6.3 |
| 103 | Markings – including 2D bar codes - missing, incorrect, illegible, incorrect size/height, or not in proper location | VL II | 5.1.2 | 4.6.4/ Gage |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL I | 3.10 | Visual |



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| 4.3.4.32 Unitization. | | | Drawing Number 19-48-4116 19-48-4116/131B (M81 Marine) 19-48-4116/131D (M81 Army) | |
|---|---|---|---|---|
| | | | Next Higher Assembly | |
| | Examination | Conformance Criteria | Requirement Paragraph | Paragraph Reference/ Inspection Method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location | VL II | 5.1 | Visual |
| 102 | Pallet load markings -including 2D bar codes - missing, incorrect, illegible, or not in proper location | VL II | 5.1.2 | 4.6.4 |
| 103 | Filler assembly not built, treated, or marked correctly | VL II | 5.1 | Visual |
| 104 | Boxes incorrectly oriented or positioned on pallet | VL II | 5.1 | Visual |
| 105 | Strapping not applied correctly or incorrect type | VL II | 5.1 | Visual |
| 106 | Unit load not tightly strapped or any strap loose | VL II | 5.1 | Visual / Manual |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL I | 3.10 | Visual |



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4.4 Test sub-procedures.

4.4.1 M700 insertion procedures. Loosen the fuse holder cap three revolutions. Loosen the plugs by pressing them into the igniter. Grip both the weatherproofing and shipping plugs and move them side to side while simultaneously pulling both plugs from the igniter. Cut a 24" ±1" length of fuse so its ends are clean and square. Insert the fuse into the igniter as far as it will go (to ensure complete and proper insertion of the fuse). Re-tighten the fuse holder cap to ensure proper holding force; finger-tight closures shall be used.

4.4.2 Shock tube insertion procedures. Loosen the fuse holder cap three revolutions. Loosen the weatherproofing plug (outermost plug) by pressing the plug into the igniter. Grip the weatherproofing plug and move it side to side while simultaneously pulling the weatherproofing plug out of the shipping plug and the igniter. Cut a 24" ±1" length of shock tube so its ends are clean and square. Seal one end of the shock tube (tape, ultrasonic, cap, etc.). Insert the unsealed end of the shock tube into the shipping plug as far as it will go (to ensure complete and proper insertion of the shock tube). Re-tighten the fuse holder caps to ensure proper holding force; finger-tight closures shall be used.

4.4.3 Functioning procedures. Remove the cotter pin, then apply an axial force to the pull ring in the direction away from the igniter. If the M81 does not function, reset per 4.4.3.1.

4.4.3.1 Quick reset procedures. To reset the igniter, reengage the striker assembly by pushing the pull rod back into the end of the igniter. Verify the pull rod re-engages the striker assembly by feeling the pull ring's increased resistance to withdrawal. After the pull rod re-engages the striker assembly, function the igniter.. If the igniter does not function after the first reset, it may be reset a second time. The M81 and the lot shall be rejected if the igniter is incapable of being reset or if it is reset twice and still does not function.

4.5 Test methods.

4.5.1 Cotter pin extraction test. This test shall be performed on the M81. Secure the igniter. Beginning at not greater than 9-lbs, apply the incremental axial force to the eye of the cotter pin (utilizing the cord) in the direction away from the igniter and cotter pin until it withdraws completely. The M81 and lot shall be rejected if one (1) or more cotter pins is withdrawn from the igniters at less than 10lbs or two (2) or more cotter pins are withdrawn from the igniters at greater than 30lbs.

4.5.2 Pull ring force limits test. This test shall be conducted on the M81. Utilizing the samples from 4.5.1 (with the cotter pin removed) secure the igniter. Beginning at not greater than 4lbs, apply an incremental axial force to the pull ring in the direction away from the igniter until it functions (as evidenced by audible noise). If the M81 does not function, reset the igniter per 4.4.3.1. The M81 and lot shall be rejected if one (1) or more samples function at less than 5lbs or greater than 20lbs, or two (2) or more samples function at greater than 15lbs but less than 20lbs.



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4.5.3 Firing pin load test. This test shall be performed on a subassembly consisting of the pull rod assembled to the striker assembly, conforming to Drawings 8822508 and 8822512 respectively. Secure the pull ring end of the pull rod. Apply an axial load not less than 15 lbs to the firing pin flange in a direction away from the pull rod. The striker assembly and pull rod and their lots shall be rejected if the roll crimp fails or the pull rod disengages.

4.5.4 Hydrogen embrittlement. This test shall be performed initially and no less than every 120 days to assure quality and process control within the zinc/chromate plating process utilized during manufacture of the firing pin spring and pull ring. Perform the test in accordance with ASTM F519 using the specified number of Type 1a.1 Standard size test specimens. If one (1) or more test fails, the parts produced using the plating/coating process since the last successful test are considered embrittling and shall be rejected.

4.5.5 Salt spray. This test shall be conducted on the pull ring, firing pin spring, and cotter pin to determine compliance with the applicable drawings' requirements. The test shall be performed in accordance with ASTM B117 and the test duration shall be 96 hours, per the drawings and referenced specifications. The frequency of the test shall be once per week. If multiple parts use the same plating process, the test need not be performed specifically during a pull ring, firing pin spring, or cotter pin plating batch; rather vary the testing batch to check the plating process across all parts. If one (1) or more samples have corrosion products (visible to the unaided eye at normal reading distance) the parts produced using the plating/coating process since the last successful test shall be rejected.

4.5.6 Attachment integrity test. This test shall be conducted on the M81. Equally divide the M81 samples into the two groups, one for insertion with time fuse and one for insertion with shock tube. Obtain lengths of M700and shock tube. Insert the lengths into M81s per 4.4.1/4.4.2. Secure the igniter and apply the specified axial force for the specified time to the junction of the M81 and length. The M81 shall be rejected if the length separates from the igniter. Carefully unsecure the igniter from the fixture. Seal the shock tube ends (tape, ultrasonic, cap, etc.). The M700 lengths may be cut to no less than a 3-inch protrusion from the igniter. Initiate the igniters per 4.4.3. Visually inspect the M700 for signs of initiation (venting, deformation, flame exiting the end). Visually inspect the shock tube for breakage of the end seal (and lack of interior explosive powder if the end seal is not broken). The M81 and lot shall be rejected if it does not initiate the time fuse/shock tube.

4.5.7 Operating temperatures. Equally divide the igniters into the two temperature groups. The items tested at Hot shall be conditioned at 125ºF +5/-0 ºF. The items tested at Cold shall be conditioned at -35ºF +0/-15ºF. Each of the two groups shall be conditioned for a not less than of 4 hours at their respective temperature. Along with the igniters, condition time fuse and shock tube, conforming to MIL-DTL-45144 and Drawing 13026126-3 respectively. From each temperature group, half of the samples shall be used to initiate time fuse and the other half shall initiate shock tube. Within 5 minutes after the removal from the chamber, the igniter shall be inspected and functioned. Insert similarly conditioned time fuse/shock tube per 4.4.1/4.4.2. The M700 lengths may be cut to no less than a 3 inch protrusion from the igniter. Initiate the igniters per 4.4.3. Visually inspect the time fuse and shock tube for signs of initiation (venting,



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deformation, flame exiting the end, etc.). The M81 and lot shall be rejected if it does not initiate the time fuse /shock tube.

4.5.8 Immersion test. The test shall be conducted on the M81. Equally divide the M81 samples into the two groups, one for immersion with time fuse and one for immersion with shock tube. Obtain lengths of M700 and shock tube, conforming to MIL-DTL-45144 and Drawing 13026126-3 respectively, and attach igniters onto both ends of the lengths per 4.4.1/4.4.2. Then submerge the entire length and attached igniters into the specified depth of water for the specified length of time. Remove the lengths and igniters from the water and within 30 minutes of removal from the water, cut the lengths midway between the igniters. The M700 lengths may be cut to no less than a 3 inch protrusion from the igniter. Seal the shock tube ends (tape, ultrasonic, cap, etc.). Initiate the igniters per 4.4.3. Visually inspect the M700 for signs of initiation (venting, deformation, flame exiting the end, etc.). Visually inspect the shock tube for breakage of the end seal (and lack of interior explosive powder if the end seal is not broken). The M81 and lot shall be rejected if it does not initiate the time fuse/shock tube.

4.5.9 Disassembly. Utilizing the functioned samples from 4.5.2, disassemble the M81 and verify that all internal components are present, properly assembled, and oriented. The M81 and lot shall be rejected if one (1) or more samples are found to have missing parts, extra parts or improperly assembled components.

4.6 Packaging inspections.

4.6.1 Leakage test. The test shall be performed on the sealed barrier bag and its contents (actual or simulated) in accordance with MIL-STD-2073-1 Leakage Test paragraph G.4.2. The test shall follow either the vacuum retention, the submersion (or immersion), or the hot water technique.

4.6.2 Heat seal seam strength test. The test shall be performed on the sealed barrier bag in accordance with MIL-STD-2073-1 Heat seal seam strength test paragraph G.4.3. The samples shall be chosen from the final heat seal seam which closes/seals the pre-assembled bag.

4.6.3 Metallic seal inspection. The metallic seal shall be inspected for presence, proper markings, and installed correctly. Pull on the metallic seal to determine if it is secure. If the seal breaks, the sample and item lot shall be rejected. If the wire breaks before the seal, the box shall be resealed using a new metallic seal.

4.6.4 Markings. Verify by visual inspection. The sample and lot shall be rejected if it does not conform to the marking requirements.

4.6.4.1 Marking resistance to water. The test shall be performed on First Article marked M2A2 ammunition can samples in accordance with the water resistance performance criteria of its respective drawing or specification (A-A-59383 paragraph 3.6 and A-A-208B paragraph 5.5, as applicable). The packaging material used shall be from lots submitted as part of First Article. The sample(s) and lot shall be rejected if the marking media is not legible, does not retain its



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color, smears, blisters, cracks, or peels. (Note: This testing is in addition to any testing required for the ink certification.)

4.6.4.2 Marking resistance to smearing. The test shall be performed on marked M2A2 ammunition can samples in accordance with the smearing resistance performance criteria of its respective drawing or specification (A-A-59383 paragraph 3.7 and A-A-208B paragraph 5.4, as applicable). The packaging material used shall be from lots submitted as part of First Article. The sample(s) and lot shall be rejected if the marking media smears. (Note: This testing is in addition to any testing required for the ink certification.)

5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with all the requirements of the drawings listed in Table III and all its subsidiary drawings.

TABLE III. Applicable M81 packaging drawings.

| ITEM | PACKAGING AND MARKING DWG(S) | SET-UP BOX DWG | BARRIER BAG DWG | OUTER PACK DWG | OVER PACK DWG | MISCELLANEOUS PACKAGING MATERIAL DWG(S) | BASIC UNITIZATION DWG 19-48-4116 AND APPENDIX |
|---|---|---|---|---|---|---|---|
| M81 (Army) | 12977192/ 12977193 | 12977189 | 12977190 | 12977191 | N/A | N/A | 19-48-4116/ 131D |
| M81 (Marine) | 12977192/ 13019660/ 13019661 | 12977189 | 12977190 | 13032542 | 7553347 | 13042427 | 19-48-4116/ 131B |

5.1.1 Barrier bag resistance to leakage. The barrier bag shall not leak as evidenced by performing MIL-STD-2073-1 Para G.4.2 (leakage test) and MIL-STD-2073-1 Para G.4.3 (heat-sealed seam test).

5.1.2 Marking ink/media. The ink and/or marking media shall be in accordance with A-A-208B, A-A-59383, or other specified specification. If a quantity is not specified in the specification, a sample of three (3) per ink/paint/surface/condition to be marked shall be used. All markings shall be smooth, uniform, clear in appearance and free of conspicuously ragged edges. There shall be no bleeding of color markings or the plastic exterior to the extent that there is a mottled or streaky appearance. The markings shall show no lifting, wrinkling, or other visible effect on the plastic's surface. The markings shall be applied on clean, dry surfaces that shall be free of dirt, oil, grease, water, corrosion, or other foreign matter. All required markings shall be present, neat, legible, only where specified, and sharply defined.



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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M81 is a weatherproof igniter intended to be used for the initiation of either time blasting fuse or shock tube. The M81 is part of the Modernized Demolition Initiators (MDI) family. The M81 is used to initiate all past and current fielded and developmental time blasting fuses and shock tubes. The M81, covered by this specification, is intended for use with military demolitions. These M81s, covered by this specification, are militarily unique as they should function under rough handling and extreme climatic and storage conditions as encountered by the United States Armed Forces.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.
b. Requirement/CDRL for Physical Configuration Audit (PCA).
c. Requirement/CDRL for First article inspection.
d. Requirement/CDRL for conformance inspection.
e. Requirement/CDRL for Ammunition Data Cards (ADC) as specified in MIL-STD-1168.
f. Requirement/CDRL for Acceptance Inspection Equipment (AIE) and associated test procedures, including M700 and shock tube to be used during testing.
g. Requirement/CDRL for Alternate Acceptance Plans.
h. Packaging requirements. Special requirements exist for the packaging of this item. Detailed directions and additional information may be obtained from FCDD-ACE-LP.
i. Requirement for certificates of findings (certificates of analysis) on the primer lot(s) to be submitted with each igniter lot.
j. Requirement for evidence that the primer is manufactured by a qualified source and the total time between original acceptance of the primer lot and its assembly into the igniter should not be greater than 2 years.

6.3 Complete functioning. The functioning of the M81 can be evidenced by the propagation of the M81's primer to the inserted time fuse or shock tube. The time fuse can be verified for propagation by looking for a flame exiting the time fuse end and/or by examining the exterior and interior, which changes in appearance when burned. The shock tube can be verified for propagation by looking for a flame exiting the shock tube and/or examining the interior walls for explosive material. If the shock tube's end is sealed (by tape, ultrasonic, cap, etc.) then a rupture of the shock tube or seal will typically occur during functioning. If the seal or shock tube



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does not rupture, the shock tube can be cut open and the interior walls inspected for the absence of explosive.

6.4 Amendment overview of changes. The following approved Engineering Change Proposals (ECPs) are being incorporated into this document: R23B2003, R235003.

6.5 Subject term (key word) listing.
Initiator
Modernized Demolition Initiators (MDI)
Pull ring

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING INFORMATION

Custodian:
Army - AR
Navy – MC
Air Force - 99

Preparing activity:
Army-AR
(Project: 1375-2023-001)

Review activities:
Navy – AS, SH
Air Force – 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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